

An end-to-end pipeline for knowledge graph population from 19th-century land registry digitised tables

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(1)



(2)



(3)



Context

- **Significant volumes of tabular documents** preserved in GLAMs
- **Information of interests in several fields** such as history, geography, economics etc.
- **Gradual digitization of these documents... but still difficult to retrieve information in them**

Examples of tabular documents archived in Archives of Val-de-Marne (French *département*)

5 866 images

Vet students registers (1760-1817)

[illegible]

24 012 images

Initial registers of the French Land Registry (1809-)

[illegible]

163 371 images

*Alphabetical tables of deaths, inheritances,
and absences (1759-1974)*

[illegible]

Research problem

How can we facilitate access to the knowledge contained in digitized historical tables?

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➡ Previous works used knowledge graphs to structure information extracted from historical documents (but not from historical tables) [1,2].

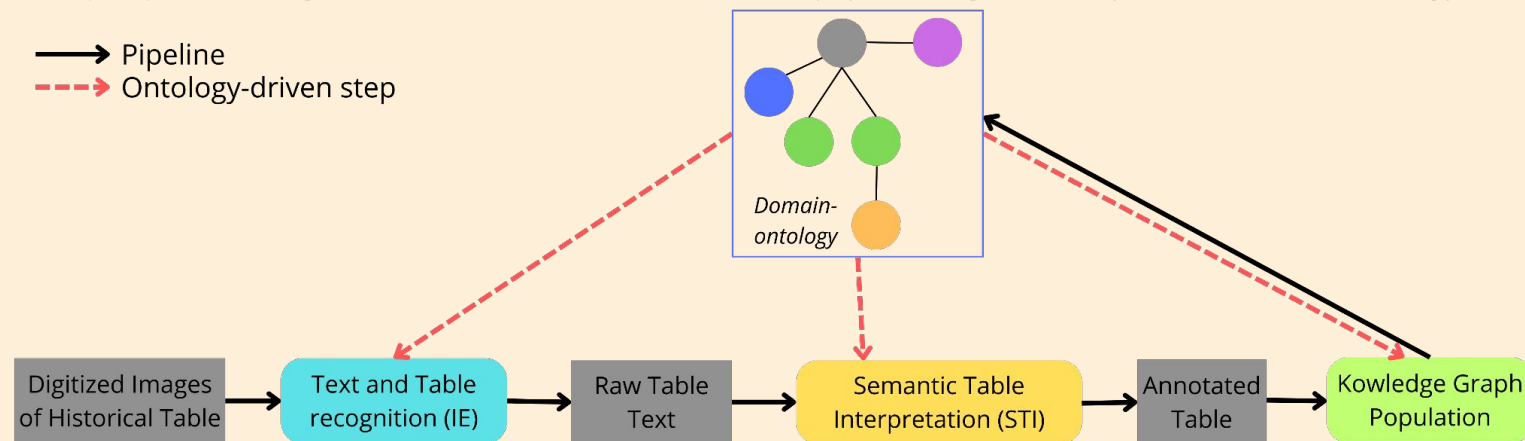
[1] V. Nundloll, et al. Automating the extraction of information from a historical text and building a linked data model for the domain of ecology and conservation science. Heliyon, 8(10) :e10710, 2022.

[2] N. Jain et al. Generating Domain-Specific Knowledge Graphs : Challenges with Open Information Extraction. In Proc. 1st Int. Workshop on Knowledge Graph Generation From Text (Text2KG 2022)

Research problem

How can we facilitate access to the knowledge contained in digitized historical tables?

➡ We propose a **digitised table-to-KG automatic pipeline guided by a domain-ontology**.



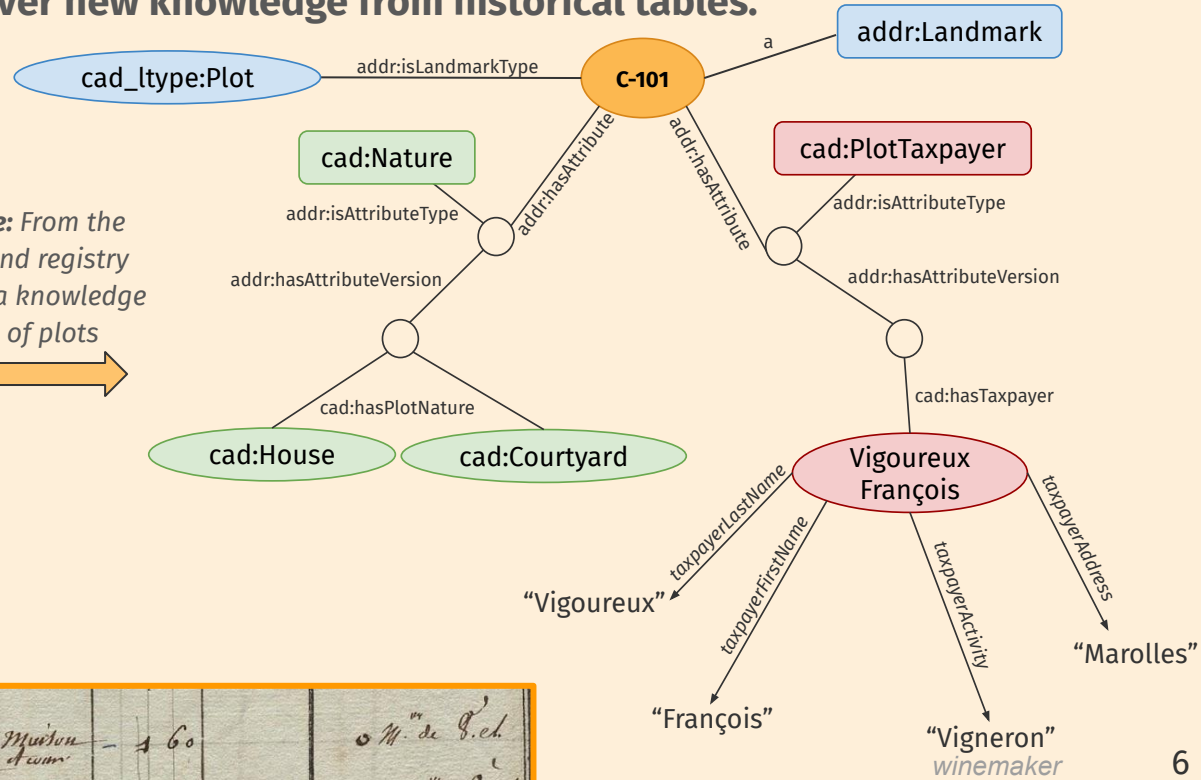
Application research question: **how to extract and structure information about land plots from the 19th-century French land registry tables?**

Semantic Interpretation of Historical Tables

Combine the Information Extraction and Semantic Table Interpretation pipelines together to structure information and discover new knowledge from historical tables.

CANTON	NUMERO	NOM	NATURE	CONTENU	PROPRIETE	REMARKS
Du Village	101	Vigoureux François Marolles	Plot	1 60		
	102	Vigoureux François Marolles	Plot	1 60		
	103	Vigoureux François Marolles	Plot	1 60		
	104	Vigoureux François Marolles	Plot	1 60		
	105	Vigoureux François Marolles	Plot	1 60		
	106	Vigoureux François Marolles	Plot	1 60		
	107	Vigoureux François Marolles	Plot	1 60		
	108	Vigoureux François Marolles	Plot	1 60		
	109	Vigoureux François Marolles	Plot	1 60		
	110	Vigoureux François Marolles	Plot	1 60		
	111	Vigoureux François Marolles	Plot	1 60		
	112	Vigoureux François Marolles	Plot	1 60		
	113	Vigoureux François Marolles	Plot	1 60		
	114	Vigoureux François Marolles	Plot	1 60		
	115	Vigoureux François Marolles	Plot	1 60		
	116	Vigoureux François Marolles	Plot	1 60		
	117	Vigoureux François Marolles	Plot	1 60		
	118	Vigoureux François Marolles	Plot	1 60		
	119	Vigoureux François Marolles	Plot	1 60		
	120	Vigoureux François Marolles	Plot	1 60		

Use case: From the initial land registry tables to a knowledge graph of plots



Du Village	101	Vigoureux François Marolles	Plot	1 60		
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19th-century French land registry tables

Studied registers : **describe the initial state of land plots in the early XIXth**

Look like relational tables... but aren't relational tables:

- Multi-rows cells
- Braced cells
- Lots of distributed values with *ditto* or blank cells distribution

Lots of **abbreviations**.



Merged Cell



Ditto (example)



Abbreviations

NOMS, PRÉNOMS, PROFESSIONS ET DEMEURES DES PROPRIÉTAIRES.	N ^{os} du PLAN rectifié.	CANTONS, TRIAGES ou LIEUX DITS.	NATURE de CULTURE.
197 E. Hilaire Louis f ^{on}	43	Saône-Des	Cer.
191 Mainfroy J ^{re} ch ^{re}	44	Postes	C
120 Barre J ^{re} Victor J ^{re}	45		C

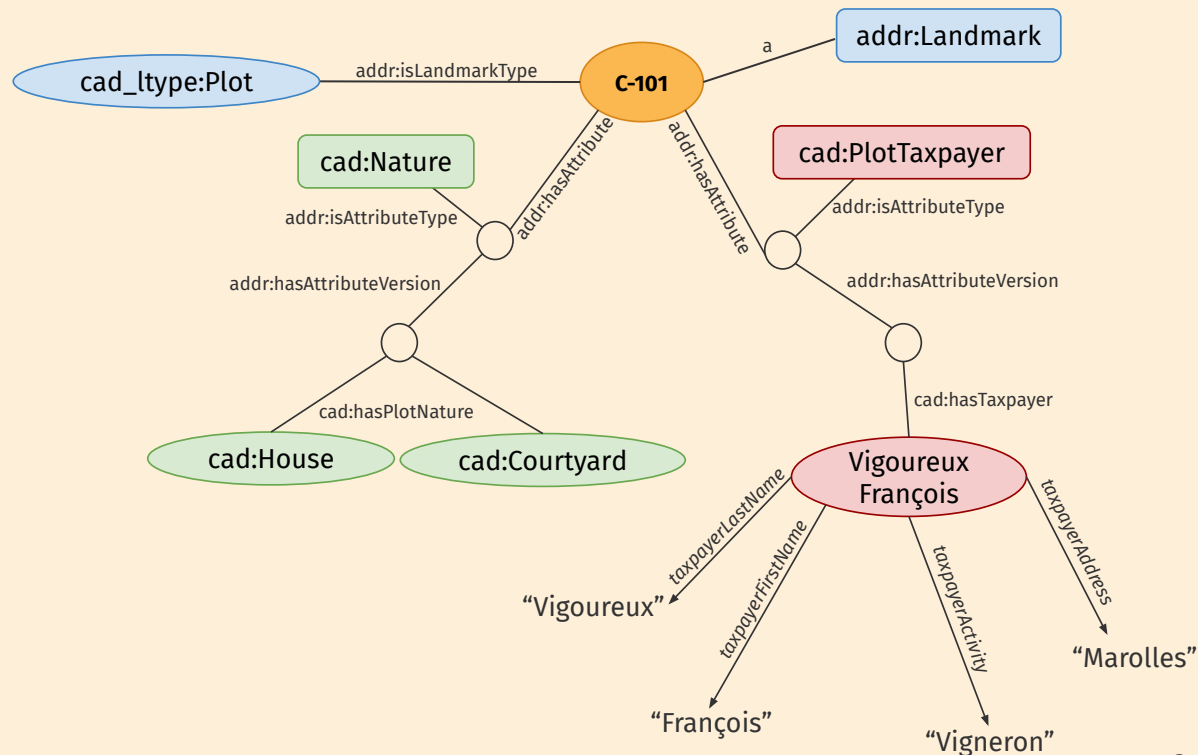
Taxp. Index in Taxpayer
next register

Plot Plot Plot
Num. Addr. Nature 7

Domaine-ontology : PeGazUs

PeGaZus : Perpetual Gazetteer of approach-address UtteranceS [1]

- Ontology to represents:
 - addresses and landmarks
 - with attributes that evolve over time
 - has an extension dedicated to plots representation



Outline

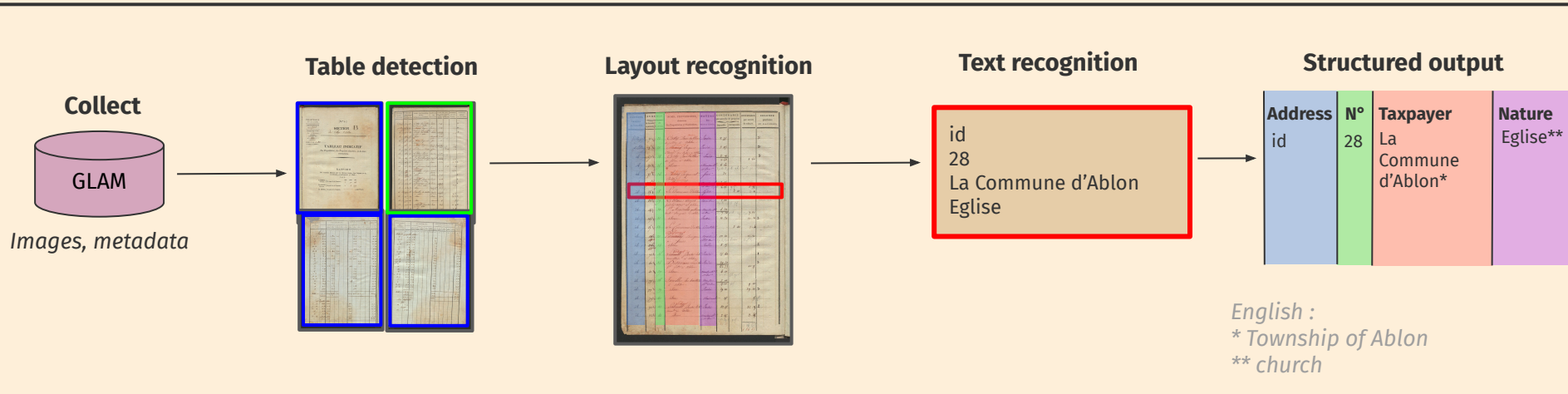
1. **State-of-the-art**
 - a. Information Extraction (IE) from historical tables
 - b. Semantic Table Interpretation (STI)
2. **Our approach for ontology population from digitized historical tables**
3. **Dataset and evaluation**
4. **Conclusion**

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Information Extraction (IE) from historical tables

- Target, transcribe, and structure the text contained in digitized tables into a structured computer-readable format.
- Main processing steps (*some might be merged in certain approaches*)
- 2 types of approaches: sequential and end-to-end



Semantic Table Interpretation (STI)

- Annotate the different elements of a table and their relationships using the entities, classes and properties of a knowledge graph (KG).

7 juin	Bâle	Suisse	0	1	Tchéquie
7 juin	Genève	Portugal	2	0	Turquie
11 juin	Genève	Tchéquie	1	3	Portugal
11 juin	Bâle	Suisse	1	2	Turquie
15 juin	Bâle	Suisse	2	0	Portugal
15 juin	Genève	Turquie	3	2	Tchéquie

Diagram illustrating Semantic Table Interpretation (STI) for a Wikipedia table about EURO 2008 football games. The table is annotated with Wikidata entities and relationships:

- CEA (Cell-Entity Annotation):** Swiss national football team (Q165141) is linked to the 'Suisse' cell.
- CTA (Column-Type Annotation):** city of Switzerland (Q54935504) is linked to the 'Bâle' and 'Genève' cells.
- CPA (Column-Property Annotation):** number of points/goals/set scored (P1351) is linked to the '0', '1', '2', '3' cells.
- Topic Annotation:** UEFA Euro 2008 (Q241864) is linked to the 'Tchéquie', 'Portugal', 'Suisse', and 'Turquie' cells.
- Row-to-instance:** Switzerland - Turkey, 11 Jun 2008 (Q12012827) is linked to the row containing '11 juin', 'Bâle', 'Suisse', '1', '2', 'Turquie'.

Wikipedia table about EURO 2008 football games annotated with Wikidata entities [1].

- Cell-Entity Annotation (**CEA**)
+ Cell New-Entity Annotation (**CNEA**)
/ Not In Lexicon (NIL) [2]
- Column-Type Annotation (**CTA**)
- Column-Property Annotation (**CPA**)
- **Topic Annotation**
- **Row-to-Instance Annotation**

Approaches developed on benchmark datasets, barely used on “real-world data” and domain-specific application.

[1] J. Liu et al. From tabular data to knowledge graphs : A survey of semantic table interpretation tasks and methods. J. Web Semant., 76 :100761, 2023.

[2] M. Cremaschi et al. Survey on Semantic Interpretation of Tabular Data : Challenges and Directions, 2024. arXiv :2411.11891

Semantic Table Interpretation (STI)

- Annotate the elements of a table and their relationships using the entities, classes and properties of a knowledge graph (KG).

cad:hasIndexNum

Plot Number 	Address 	Taxpayer 	Taxpayer Mutation Register ID 	Plot Nature 
26	Village d'Ablon	<i>houry</i> Amy Jacques	72 ^{bis}	Four
27	id	<i>houry</i> Amy Jacques	72 ^{bis}	Maison B ^t et Cour
28	id	La Commune d'ablon	25	Eglise

addr:Landmark entity **cad:Taxpayer** **pnature:Church**

- Cell-Entity Annotation (**CEA**)
+ Cell New-Entity Annotation (**CNEA**)
/ Not In Lexicon (NIL) [2]
- Column-Type Annotation (**CTA**)
- Column-Property Annotation (**CPA**)
- **Topic Annotation**
- **Row-to-Instance Annotation**

?row a addr:Landmark;
addr:isLandmarkType cad_ltype:Plot.

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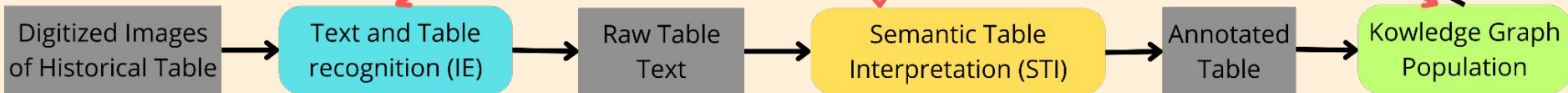
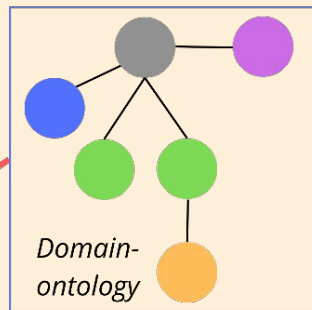
Semantic Interpretation of Historical Tables

Combine the IE and STI pipelines together to structure information and discover new knowledge from historical tables.

➡ All steps are driven by the domain ontology!

→ Pipeline

- - -> Ontology-driven step



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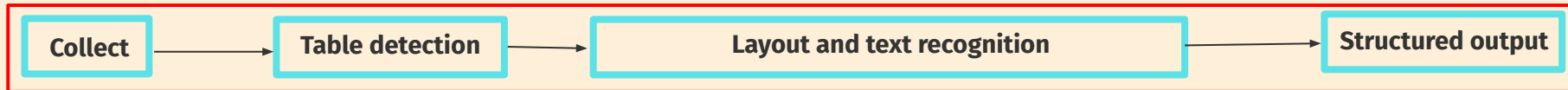
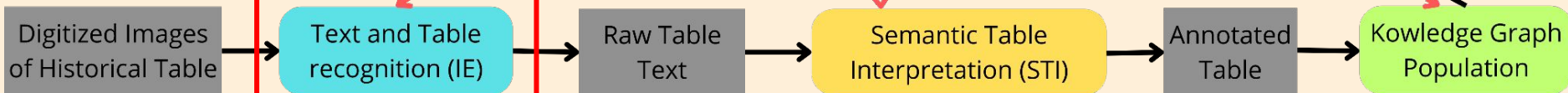
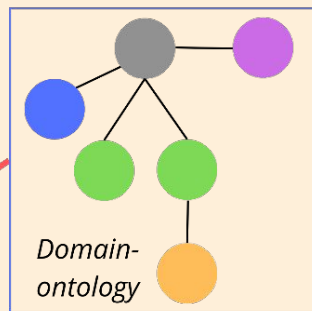
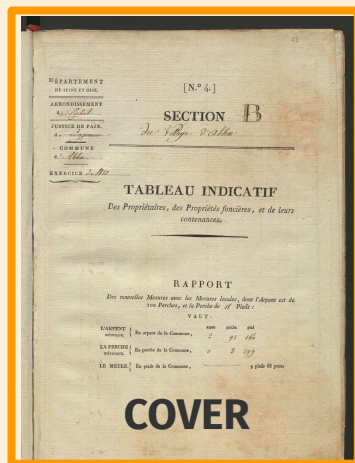


Table detection

Pages classification
(for this use case)
using YOLOv11-cls.

The classification step
is required to filter the
different types of
pages and **identify the
tables.**

Initial registers,
Marolles-en-Brie, 1810.



COVER

TABLE

TABLE

CLASSEMENT

RECAPITULATION DES CONTENANCES

RELEVÉ des Nombres de Mesures

CLASSEMENT

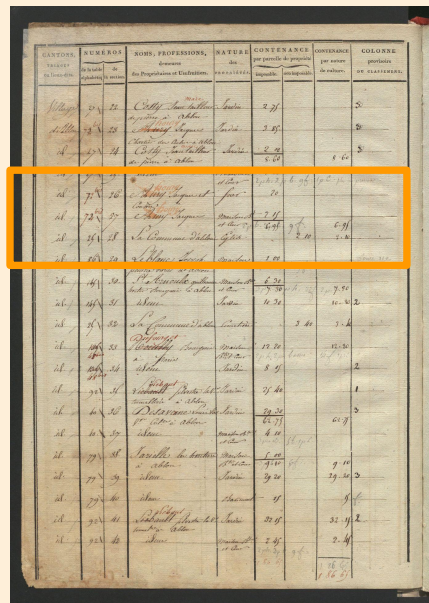
CLASSEMENT

RECAPITULATION de la contenance de la commune

PROPRIÉTÉS BATIES

Information extraction from historical tables

- Page-level information extraction approach
 -  one unique dataset (image table → textual table) : **Document Attention Network (DAN)** [1]
 -  deals with layout variations (3 types of initial registers tables in dataset)
 -  *full page* context provided for each row processing
 -  no error propagation from layout detection to text recognition
 - !! hallucinations



Text and
layout
recognition



Plot Number 	Address 	Taxpayer 	Taxpayer Mutation Register ID 	Plot Nature 
..	..	..	..	..
27	id	<i>houry</i> Amy Jacques	72 ^{bis}	Maison B ^t et Cour
28	id	La Commune d'ablon	25	Eglise
..	..	..	..	..

Information Extraction from digitised tables

- **Page-level end-to-end IE approach**
- Model: Document Attention Network (DAN) [1]
 - Fine-tuned with a 139 pages manually annotated dataset (97/22/28)
- Global CER on test set (28 pages of various qualities): 35,3%

NOMS. PRÉNOMS, PROFESSIONS ET DEMEURES DES PROPRIETAIRES.	N° du PLAN rectifié.	CANTONS, TRIAGES ou LIEUX DITS.	NATURE de CULTURE.
297 Thibault Louis f [↑] ois [↓]	43	La Voie des→	Terre
201 Mainfray p [↑] re [↓] ch [↑] es [↓]	44	←Postes	T
20 Barrué j [↑] ques [↓] victor	45	§	T

Typography and layout explanation

 Merged Cell
 Ditto
 ↑ Exponent

Additional typography and layout tokens

§ ditto
 ↑ exponent ↓
 go to→the line absent here
to→←merge
×crossed-out± absent here

Ground-truth text

297	Thibault louis f ↑ ois ↓	43	La Voie des→	Terre
201	Mainfray p ↑ re ↓ ch ↑ es ↓	44	←Postes	T
20	Barrué j ↑ ques ↓ victor	45	§	T

Predicted text

297	Thibault louis f ↑ ois ↓	43	La Voie des→	Terre
201	Mainfray p ↑ re ↓ ch ↑ es ↓	44	§	T
20	Bar m é j ↑ ques ↓ victor	45	§	T

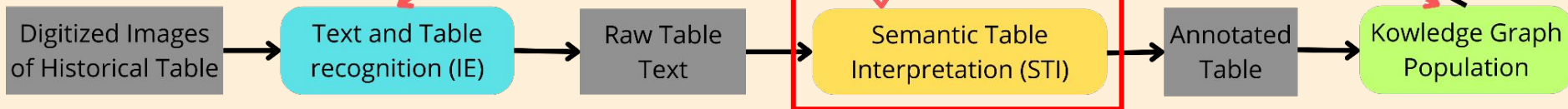
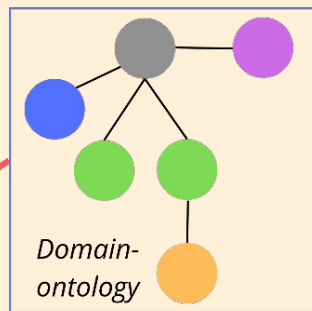
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Semantic interpretation of historical tables

- IE annotation model refers to the domain ontology $\Rightarrow$ solves some STI tasks.

Before 1822

CANTONS, TRIAGES ou lieux-dits.	NUMÉROS de la table alphabétique section.	NOMS, PROFESSIONS, demeures des Propriétaires et Unfruitiers.	NATURE des PROPRIÉTÉS.	CONTENANCE par parcelle de propriété par nature de culture.	provisoire DU CLASSEMENT.
Chammi	95. 166	Lefèvre	terres	275.10	1
Chammi	25. 167	Deschamps, Pierre	terres	1135	1
	29. 168	Caron.	terres	9410	1
	95. 169	Lefèvre	terres	2590	1
	25. 170	Deschamps, Pierre	terres	0720	1
	25. 171	Leves	terres	0540	1
	81. 172	Leves	terres	0615	1
	81. 173	Leves	terres	0610	1

After 1822

NOMS, PRÉNOMS, PROFESSIONS ET DEMIÈRES DES PROPRIÉTAIRES.	N° PLAN.	CANTONS ou lieux-dits.	NATURE des PROPRIÉTÉS.	CONTENANCE.	CLASSES.	REVENU.
Leves, Denis Vincent	158. 159	Leves	terres	1. 10	1. 10	1. 10
Leves, Jacques	158. 160	Leves	terres	1. 10	1. 10	1. 10
Leves, Pierre	158. 161	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 162	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 163	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 164	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 165	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 166	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 167	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 168	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 169	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 170	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 171	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 172	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 173	Leves	terres	1. 10	1. 10	1. 10

Second land registry of Seine dept.(1835-1845)

NOMS, PRÉNOMS, PROFESSIONS ET DEMIÈRES DES PROPRIÉTAIRES.	N° PLAN.	CANTONS, TRIAGES ou lieux-dits.	NATURE des PROPRIÉTÉS.	CONTENANCE en ares.	CLASSES.	REVENU.
Leves, Denis Vincent	158. 159	Leves	terres	1. 10	1. 10	1. 10
Leves, Jacques	158. 160	Leves	terres	1. 10	1. 10	1. 10
Leves, Pierre	158. 161	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 162	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 163	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 164	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 165	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 166	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 167	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 168	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 169	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 170	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 171	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 172	Leves	terres	1. 10	1. 10	1. 10
Leves, Jean	158. 173	Leves	terres	1. 10	1. 10	1. 10

Here: “PERpetual GAZeteer of approach-address UtteranceS” ontology (PeGazUs) [1]

Column-Type Annotation

addr:Landmark	cad:Taxpayer
xsd:string	cad:Nature
xsd:string	xsd:string

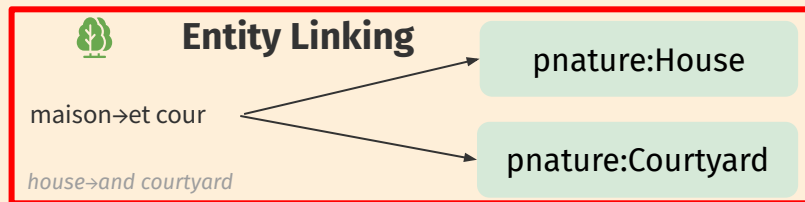
Column-Property Annotation: solved by the domain-ontology

Row type

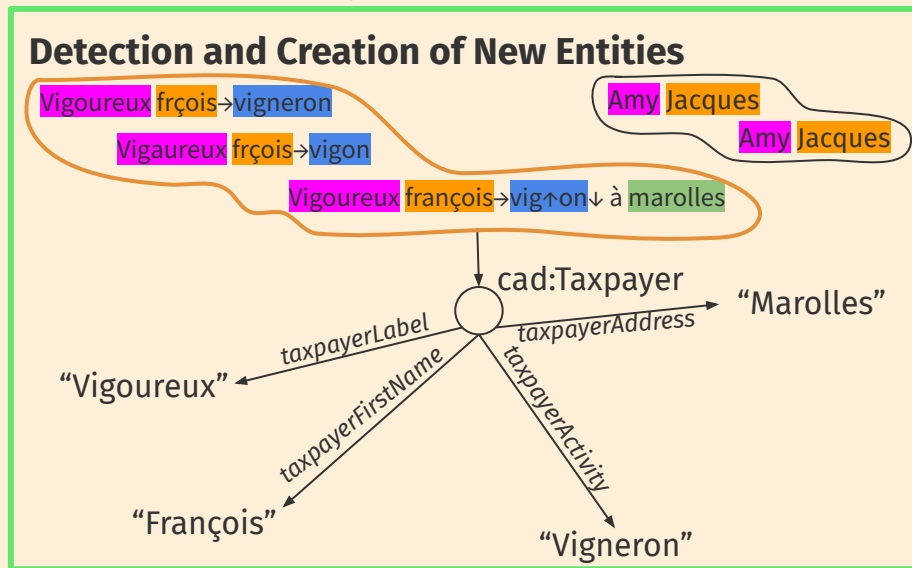
Here : ?row a add:Landmark;
add:isLandmarkType cad:Plot

Semantic interpretation of historical tables

- Dedicated processings to solve the remaining tasks for each column type: Cell-[New] Entity Annotation
 - **CEA**: compute “Cell Values” with “Entities” labels similarity and set links between them.
 - NIL value is a possible result
 - **CNEA**: [NER +] clustering based process to create **new entities**
 - NER is optional. Performed for “Taxpayers” with a fine-tuned CamemBERT model [1]
 - Clustering based on Cell Values/Named Entities similarity



Example of “Nature”



Example of “Taxpayer” : François Vigoureux, wine producer in the town of Marolles (France)

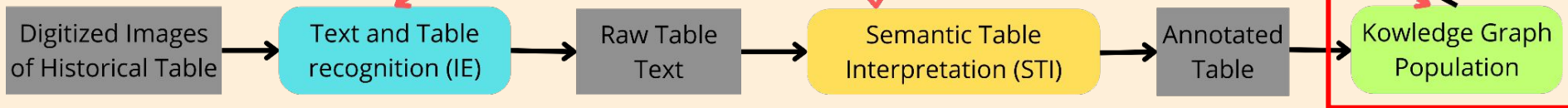
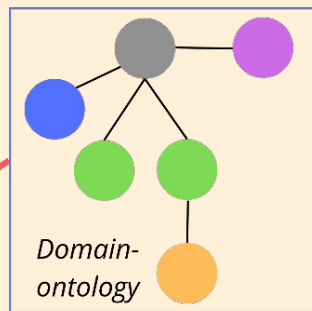
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→ Pipeline

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Knowledge graph population

- Populate the PeGazUs ontology with
 - **Entities and relations discovered in the tables**
 - **Entities and relations built using metadata**
 - Describe source documents
 - Preserve link between knowledge and their source : use the Records in Context Ontology (RiC-O) [1]

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Evaluation dataset

- Regarding CER per page on DAN test set:
 - Select a town with various CER values for its pages = **representative of outputs on the IE testset**
 - 28 pages
 - XX plots on the map
- Create a **gold standard KG** for an area of that town
- **Despite the variable IE scores (as seen on DAN test set), are we able to create the entities/properties of the KG ?**

Example of page from our evaluation dataset.
Yellow boxes are DAN attention areas associated with each row.

SOMM. PREVIOUS.	N°	CAPIT. TITRES	DATE	CONTENU	POPULATION DES NOMBRES	REMARQUES	REMARQUES
PROPRIETES ET DEMURES	DE	DE	DE	DE	DE LA COMMUNE, DE LA CANTON, DE L'ARRONDISSEMENT	DE LA COMMUNE, DE LA CANTON, DE L'ARRONDISSEMENT	DE LA COMMUNE, DE LA CANTON, DE L'ARRONDISSEMENT
DES PROPRIETES	DE	DE	DE	DE	DE LA COMMUNE, DE LA CANTON, DE L'ARRONDISSEMENT	DE LA COMMUNE, DE LA CANTON, DE L'ARRONDISSEMENT	DE LA COMMUNE, DE LA CANTON, DE L'ARRONDISSEMENT
1. Broussin, p. 1	1	1	1	1	1	1	1
2. Broussin, p. 2	2	2	2	2	2	2	2
3. Broussin, p. 3	3	3	3	3	3	3	3
4. Broussin, p. 4	4	4	4	4	4	4	4
5. Broussin, p. 5	5	5	5	5	5	5	5
6. Broussin, p. 6	6	6	6	6	6	6	6
7. Broussin, p. 7	7	7	7	7	7	7	7
8. Broussin, p. 8	8	8	8	8	8	8	8
9. Broussin, p. 9	9	9	9	9	9	9	9
10. Broussin, p. 10	10	10	10	10	10	10	10
11. Broussin, p. 11	11	11	11	11	11	11	11
12. Broussin, p. 12	12	12	12	12	12	12	12
13. Broussin, p. 13	13	13	13	13	13	13	13
14. Broussin, p. 14	14	14	14	14	14	14	14
15. Broussin, p. 15	15	15	15	15	15	15	15
16. Broussin, p. 16	16	16	16	16	16	16	16
17. Broussin, p. 17	17	17	17	17	17	17	17
18. Broussin, p. 18	18	18	18	18	18	18	18
19. Broussin, p. 19	19	19	19	19	19	19	19
20. Broussin, p. 20	20	20	20	20	20	20	20
21. Broussin, p. 21	21	21	21	21	21	21	21
22. Broussin, p. 22	22	22	22	22	22	22	22
23. Broussin, p. 23	23	23	23	23	23	23	23
24. Broussin, p. 24	24	24	24	24	24	24	24
25. Broussin, p. 25	25	25	25	25	25	25	25
26. Broussin, p. 26	26	26	26	26	26	26	26
27. Broussin, p. 27	27	27	27	27	27	27	27
28. Broussin, p. 28	28	28	28	28	28	28	28

STI and KG evaluation

- Comparison of the gold standard KG and the automatically built KG
 - **Agreement, deficit and surplus rates** $\Leftrightarrow$ created entities and properties

	Agreement	Deficit	Surplus
Instances	87.26%	12.74%	11.94%
Relations (object type properties)	78.5%	21.5%	19.9%
Datatype properties (without their values)	87.35%	12.65%	16.66%

- Agreement : instances/properties retrieved in both graphs
- Deficit : % of instances/properties missing in the automatically built graph
- Surplus : % of instances/properties wrongly added in the automatically built graph

STI and KG evaluation

Instances by class evaluation

Use of metadata only
(✓✓/✓)

Rely on raw detection
by DAN (✓)

Created with complex
cells OR Rely on other
instances that have
been miscreated
(—/—)

Type	Agreement	Deficit	Surplus	Actual	Predicted
addr:Event	100	0	0	1	1
rico:RecordSet	100	0	0	2	2
rico:Record	100	0	0	30	30
rico:Instantiation	95.44	4.56	3.33	395	390
addr:Attribute	95.20	4.80	3.43	1084	1050
addr:AttributeVersion	95.20	4.80	3.43	1084	1050
addr:Landmark	94.28	5.72	7.24	367	373
addr:Change	93.91	6.09	3.48	1446	1407
cad:Taxpayer	60.53	39.47	50.27	152	185
addr:LandmarkRelation	47.39	52.61	51.54	728	712

Conclusion

- End-to-end pipeline for the automatic extraction and semantic structuring of the information from 19th-century French land registry tables.
 - **IE steps + STI steps: A domain-ontology to rule them all ! (PeGazUs ontology)**
- Evaluation conducted on a set of tables that are geographically consistent.
- Future work:
 - Extend IE dataset to increase IE model performances (DAN)
 - Extend the gold standard graph to produce an extended evaluation dataset
 - Extend the pipeline to the mutation registers (more complexe tables)
 - Make the KG accessible from the source images (IIIF API)



Paper with appendices, datasets, code, models:
https://github.com/solenn-tl/land_registry_tables_processing_TPD_L_2025

References

- M. **Boillet**, S. Tarride, Y. Schneider, B. Abadie, L. Kesztenbaum, and C. Kermorvant. The Socface Project: Large-Scale Collection, Processing, and Analysis of a Century of French Censuses. In ICDAR, pages 57–73, 2024.
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- J. **Liu**, Y. Chabot, R. Troncy, V.-P. Huynh, T. Labbé, and P. Monnin. “From tabular data to knowledge graphs: A survey of semantic table interpretation tasks and methods.” J. Web Semant., 76: 100761, 2023.

Application context

The 19th-century French land registry

Purpose: Administrative tool to rule property taxes, enhancing efficiency and fairness.

Components:

- **Maps:** Used to locate land plots.
- **Registers :** Provide detailed descriptions of the plots.

Availability: Most towns have at least one version of the land registry created between 1808 and 1845.

Time cover: Used until land registry renovation.

Renovated land registry introduced between 1930 and 1975, varying by town.

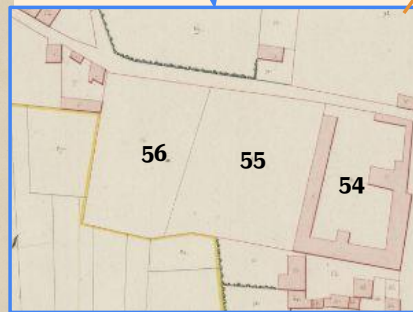
Critical analysis of the 19th-century French land registry tables

Focus on the initial registers:

Assembly map of the cadastral index maps (1810)



Cadastral index map (1810)



Town of Marolles-en-Brie,
Section C

Initial register (1812)

CANTON, TRIAGES, ou Lieux dits.	NUMÉROS DU PLAN provisions distinctes.	NOMS, PROFESSIONS, DE MEURES des Propriétaires ou Usultiers.	NATURES des Propriétés.	CONTENANCE par Propriété.	CONTENANCE par nature de culture.
	+ 54	Beuthier, Alexandre, Propriétaire	Maisons et Cour	30 15	0. M. 2. 1/2
	+ 55	Idem	Jardin	37 60	37 60 2 ^e
	+ 56	Idem	Vignes, etc.	29 00	29 00 1 ^e

Initial description of plots C-54, C-55 and C-56

Mutation registers (valid from 1812 to 1822)

ARTICLES.	NOMS, PRÉNOMS, PROFESSIONS ET DE MEURES des Propriétaires et Usultiers.	INDICATION				NOMBRE D'ARPENS, PERCHES ET MÈTRES,		CLASSES	REVENU		RENVOLS pour les Mutations,	
		de la section.	du N° de la section.	du Canton, triage ou lieu dit.	de la nature de la propriété.	par article de section.	par article de MATRICE.		de chaque article de section.	de chaque article de MATRICE.	tiré de l'article.	posté à l'article.
3	Beuthier, Alexandre, Limon, Barthelemy, et Fils, Propriétaires					arp. p. m.	arp. p. m.		fr. c.	fr. c.		
		B	90	Beffesard, Seine		2 06 1/2		2	94 3/4			
						0 11 7/8		2	11 00			
						0 30 1/2		1	19 60			
						0 37 60		2	29 70			
						0 29 00		1	24 67			

States of plots C-54, C-55 and C-56 at a given time in the first mutation register

Critical analysis of the 19th-century French land registry tables

- Several table layouts observed

Here: Table structures of initial registers from Seine and Seine-et-Oise departments (near Paris).

Before 1822

CANTONS, TRIAGES ou lieux-dits.	NUMÉROS de la table alphabétique de section.	NOMS, PROFESSIONS, demeures des Propriétaires et Usufruitiers.	NATURE des PROPRIÉTÉS.	CONTENAN par parcelle de propriété impossible. ou impossible.	par nature de culture.	provisoire DU CLASSEMENT.
Chemin de Brie	95. 186	Lefevre	terres	275.10	1	1
Chemin de Brie	25. 187	Bouillon (pauvre)	terres	1125	1	1
	29. 188	Caron.	terres	96.10	1	1
	95. 189	Lefevre	terres	2590	1	1
	25. 190	Bouillon, pauvre	terres	07.20	1	1
	25. 191	Levan	terres	05.10	1	1
	81. 192	Levan	terres	06.55	1	1
	81. 193	Levan	terres	06.10	1	1

After 1822

NOMS, PRENOMS, PROFESSIONS ET DEMIERS DES PROPRIÉTAIRES.	N° de plan.	CANTONS ou lieux-dits.	NATURE des PROPRIÉTÉS.	CONTENANCE.	CLASSE.	REVENU.
Levan, Louis Vincent	95. 186	La Ferté	terres	275.10	1	1
Melenc, gascron guillaume	95. 187		terres	1125	1	1
Caron, Louis	95. 188		terres	96.10	1	1
Bouillon, Louis	95. 189		terres	2590	1	1
Lefevre, Louis	95. 190		terres	07.20	1	1
Levan, Louis	95. 191		terres	05.10	1	1
Levan, Louis	95. 192		terres	06.55	1	1
Levan, Louis	95. 193		terres	06.10	1	1

Second land registry of Seine dept. (1835-1845)

NOMS, PRENOMS, PROFESSIONS ET DEMIERS DES PROPRIÉTAIRES.	N° de plan.	CANTONS, TRIAGES ou lieux-dits.	NATURE des PROPRIÉTÉS.	CONTENANCE.	CLASSE.	REVENU.
Levan, Louis Vincent	95. 186	La Ferté	terres	275.10	1	1
Melenc, gascron guillaume	95. 187		terres	1125	1	1
Caron, Louis	95. 188		terres	96.10	1	1
Bouillon, Louis	95. 189		terres	2590	1	1
Lefevre, Louis	95. 190		terres	07.20	1	1
Levan, Louis	95. 191		terres	05.10	1	1
Levan, Louis	95. 192		terres	06.55	1	1
Levan, Louis	95. 193		terres	06.10	1	1

Named place where the plot is located

Plot number

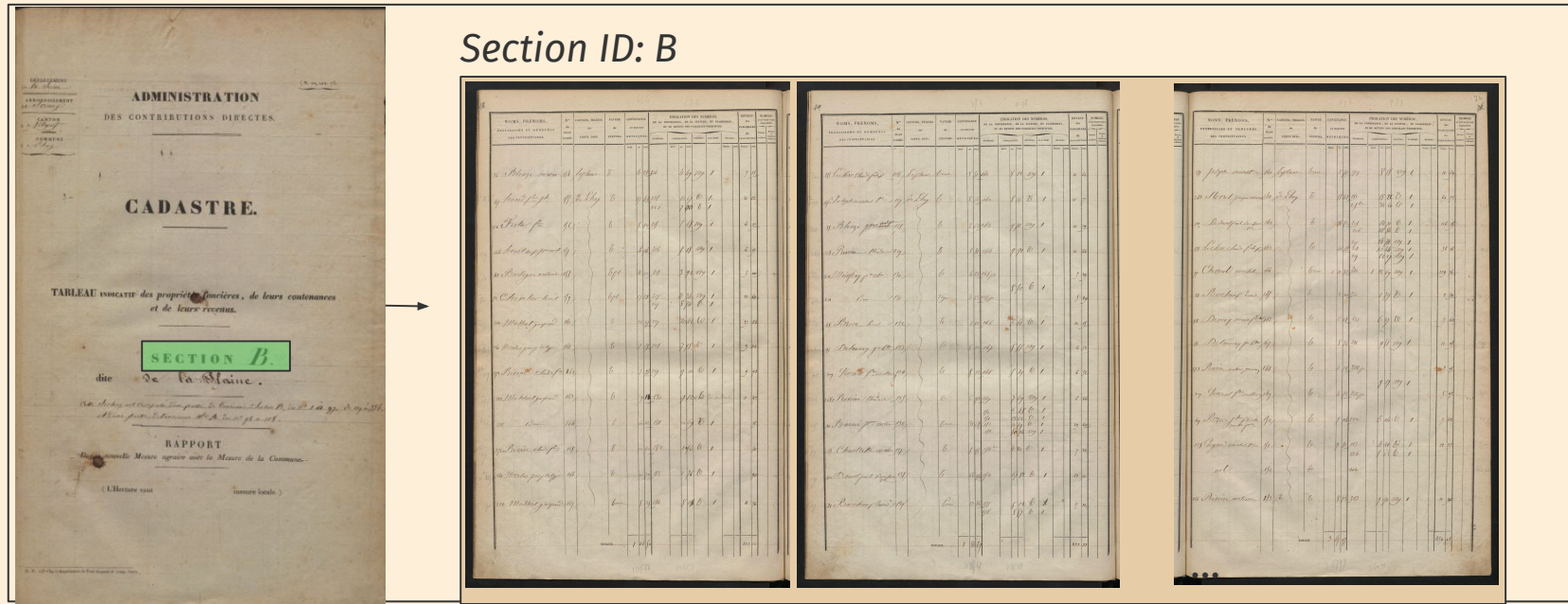
Taxpayer index number in the mutation register

Taxpayer identity

Plot nature

Critical analysis of the 19th-century French land registry tables

- Additional, required, contextual information:
 - in the images metadata
 - on the cover page of the chapter where the table is located (one chapter by part of the town)



Critical analysis of the 19th-century French land registry tables

Observations:

- Pseudo-relational tables (e.g.: merged cells)
- Various layouts
- *Dittos* to make explicits
- Abbreviations in text
- Contextual information: metadata, other pages of the register

Requirements:

- An Image-to-text information extraction process that preserves layout information
- An IE output process to turn raw text tables to knowledge (RDF resources)

Context

- **Significant volumes of tabular documents** preserved in GLAMs
- **Information of interests in several fields** as history, geography, economics etc.
- **Gradual digitization of these documents...**
 - Distribution of images on the Web
 - The information they contain are not searchable (in most of cases)

 **Documents that remain difficult to exploit (large volumes).**

Sequential approaches

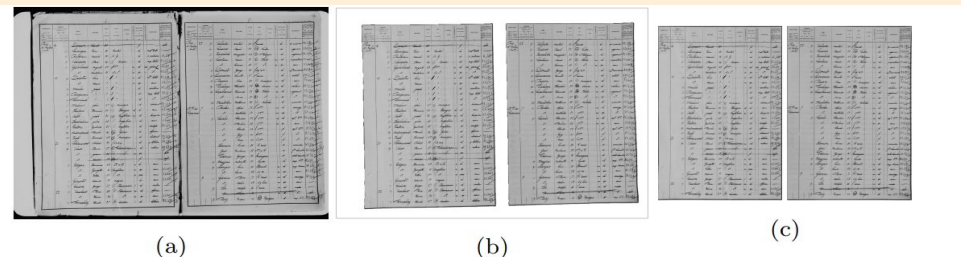
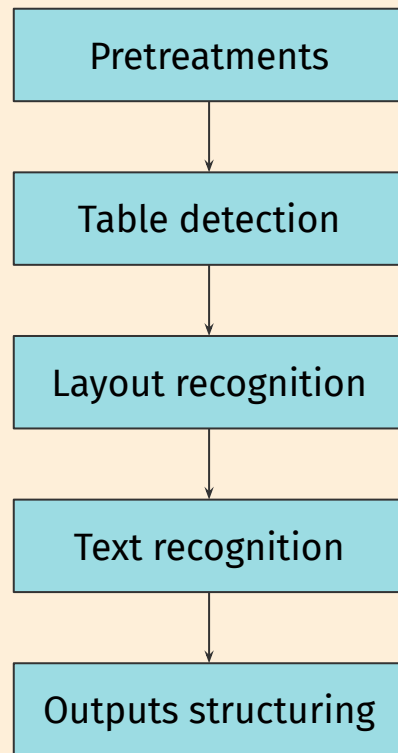


Fig. 3: (a) Example of a double page from the 1926 census. (b) Segmentation results (c) Dewarped tables.

DESIGNATION		NUMEROS PAR QUARTIER, VILLAGE, HAMEAU ou FOC			NOMS DE FAMILLE	PRENOMS	ANNEE de NAISSANCE	LIEU de NAISSANCE	NATIONA- LITE	ETAT MATRIMONIAL PAR RAPPORT au chef de ménage	SITUATION DEGRÉ D'INSTRUCTION	PROFESSION	15-352
des QUAR- TIERs, villages ou hameaux	des RUES dans les villes	des maisons	des ménages	des individus									
1	2	3	4	5	6	7	8	9	10	11	12	13	14
rué Bisson (ville)		N° 47 (link)			Cathelain	Louis	81	Aube		ch.		manoeuvre	15-352
					May	Hortense	74	Aube		Y	A		16-352
					Sarguissian	Caroizian	05	Charguie		ch.	?	s. p. 33/108	
					Tournival	Henri	96	S.		ch.		g. de café	15-

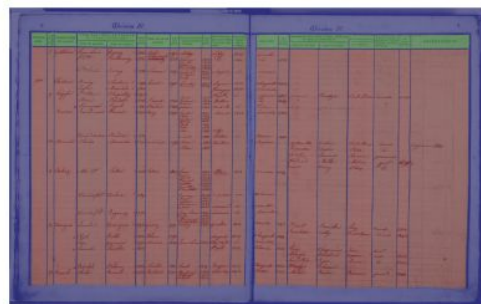
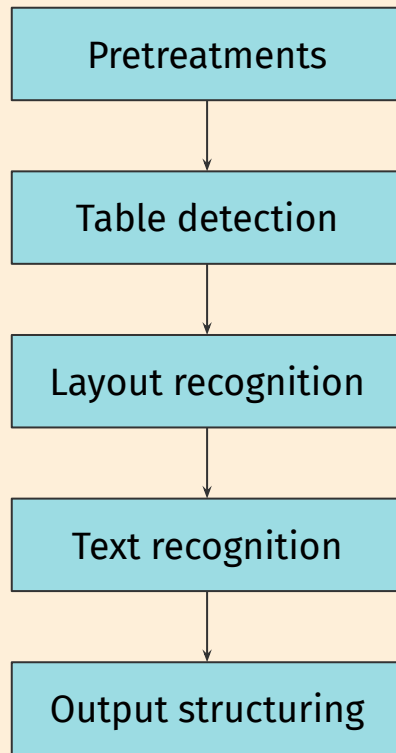
Fig. 1: Headers and first lines of a table from 1926. The corresponding annotation of the first line is "Cathelain/Louis/81/Aube/□/□/ch./□/manoeuvre/?16-352".

[1]

[1] T. Constum et al. Recognition and Information Extraction in Historical Handwritten Tables : Toward Understanding Early 20th Century Paris Census. In Document Analysis Systems, pages 143–157, 2022.

[2] R. Petitpierre et al. An end-to-end pipeline for historical censuses processing. IJDAR, 26(4) :419–432, 2023.

Sequential approaches



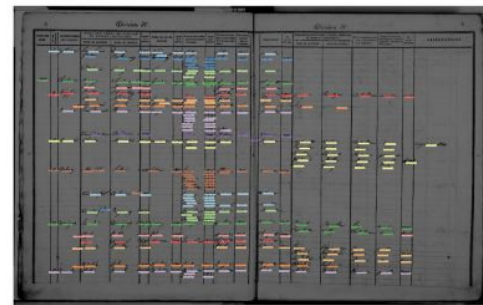
(a) Semantic segmentation of the structure



(b) Columns identification



(c) Detection of the text segments



(d) Household attribution

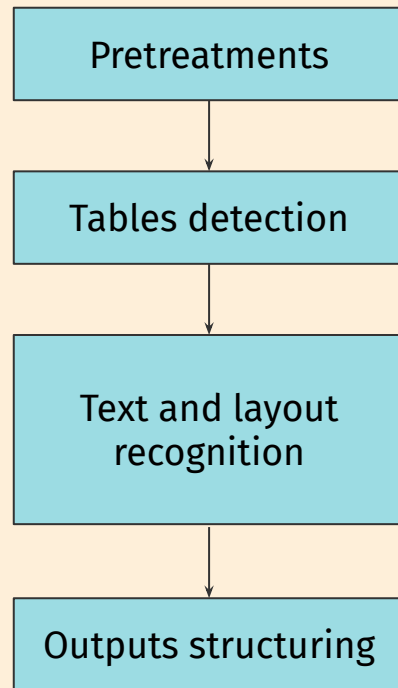
Fig. 3 Results from the main steps of the extraction pipeline on the second page of the 1832 census

[2]

[1] T. Constum et al. Recognition and Information Extraction in Historical Handwritten Tables : Toward Understanding Early 20th Century Paris Census. In Document Analysis Systems, pages 143–157, 2022.

[2] R. Petitpierre et al. An end-to-end pipeline for historical censuses processing. IJDAR, 26(4) :419–432, 2023.

End-to-end approaches



DAN [2],
DANIEL [3]

DÉSIGNATION		NUMÉROS			NOMS DE FAMILLE	PRÉNOMS	ÂGE	NATIONALITÉ	SITUATION PAR RAPPORT au chef de ménage	PROFESSION	Pour les patrons, chefs d'entreprise, ouvriers à domicile, inscrivez patrons. Pour les employés et ouvriers, indiquez le nom du patron ou de l'entreprise qui les emploie.
des quartiers villages ou hameaux	des maisons dans les villages	des maisons	des ménages	des individus							
1	2	3	4	5	6	7	8	9	10	11	12
Neuilly-le-Réal Coulgny	411	58	1861	Gendre	Pierre	75	fr.	chef	cult.	pat.	
		1862	Paraud	Marie	66	2	épouse	néant	néant	néant	
		1863	Martin	Pierre	69	7	chef	métayer	pat.	pat.	
		1864	Joyoz	Suzanne	72	2	mère	néant	néant	néant	
		1865	Martin	André	38	3	chef	métayer	pat.	pat.	

Fig. 5: Table header and first rows of a table from the census of the commune of Neuilly-le-Réal (department of Allier) in 1901. The label used to train the model for this part of the table is:

```

<s-h>Gendre <f>Pierre <o>cultivateur <l>chef <e>patron <a>75 <n>française
<s>Paraud <f>Marie <o>néant <l>épouse <e>néant <a>66 <n>idem
<s-h>Martin <f>Pierre <o>métayer <l>chef <e>patron <a>69 <n>idem
<s>Joyoz <f>Suzanne <o>néant <l>mère <e>néant <a>72 <n>idem
  
```

[1]

[1] M. Boillet et al. The Socface Project : Large-Scale Collection, Processing, and Analysis of a Century of French Censuses. ICDAR, pages 57–73, 2024.

[2] D. Coquenat, C. Chatelain, T. Paquet. DAN : a Segmentation-free Document Attention Network for Handwritten Document Recognition”. IEEE Transactions on Pattern Analysis and Machine Intelligence

[3] T. Constum. DANIEL : a fast document attention network for information extraction and labelling of handwritten documents. IJDAR, pages 1–23, 2025.

Information Extraction (IE) from historical tables : takeaways

- Two types of approaches, using **deep neural network models**

	Sequential approaches	End-to-end approaches
Explicability	+ (error propagation)	—
Dataset requirements	1 for each task requiring training	1 dataset to train models for several tasks
Use of context	—	+ (if page-level extraction)



End-to-end approaches, mostly using Transformers, allow **page level extraction**.



Still lots of challenges to extract tables with complex structures (merged/propagated cells, multiple text orientations)

NOMS, PRÉNOMS, PROFESSIONS ET DEMEURES des Contribuables, Propriétaires, Détenteurs et Vendeurs.	Noms des Propriétaires foncier.	N° de la parcelle.	NOMS des Villages, Fermes, Cultures ou Trains.	NATURE des parcelles.	CONTENU en mesures carrées.	CLASSEMENT des parcelles.	REVENUS	NOMBRE D'IMPÔTES DÉCLARÉS			
								Propriétaires	Propriétaires foncier	Propriétaires foncier	Propriétaires foncier
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/14	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/15	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/16	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/17	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/18	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/19	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/20	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/21	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/22	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/23	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/24	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/25	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/26	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/27	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/28	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/29	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/30	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/31	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/32	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/33	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/34	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/35	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/36	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/37	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/38	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/39	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/40	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/41	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/42	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/43	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/44	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/45	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/46	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/47	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/48	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/49	Ponguen	ter	1/11/2	ter	1/1				
Ponguen 060ran V↑e↓ et ent Paul 0194 Otat	Ponguen	170/50	Ponguen	ter	1/11/2	ter	1/1				

0Ponguen 060ran V↑e↓ et ent Paul 0194 Otat

STI Approaches

Heuristic approaches

Similarity measures, TF-IDF, voting, probabilities, etc.

AND

Candidates reranking

Ex : table cell values and entities labels similarity measures to perform CEA

Feature engineering

Feature extraction

AND

Machine learning models training (Random Forest, SVM, K-NN)

Ex : use column cells characteristics to predict column types (CTA)

Deep learning

Use of embedding-based models to compute similarities between tables and graphs features.

Requirement : training datasets

Semantic Interpretation of Tables : takeaways

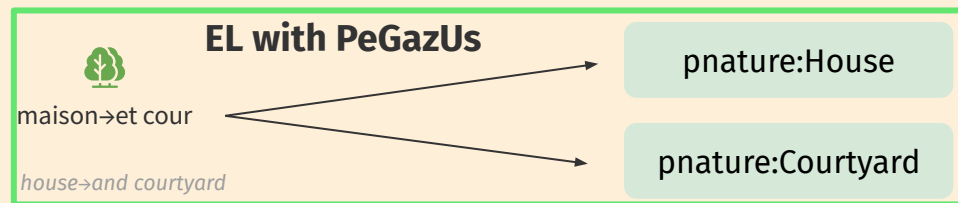
- **Multiple works to annotate relational or entity tables**
 - Tables with complex structures, hidden or composite subjects ?
- **STI often performed with encyclopedic knowledge graphs**
 - Domain-specific knowledge graphs?
- **Evaluations conducted with state-of-the-art datasets**
 - Use cases on "real" data ?
- **Not all approaches consider NIL (Not In Lexicon) entities**
 - Essential for historical applications
- **Underuse of table context**
 - Context is very important for understanding historical tables



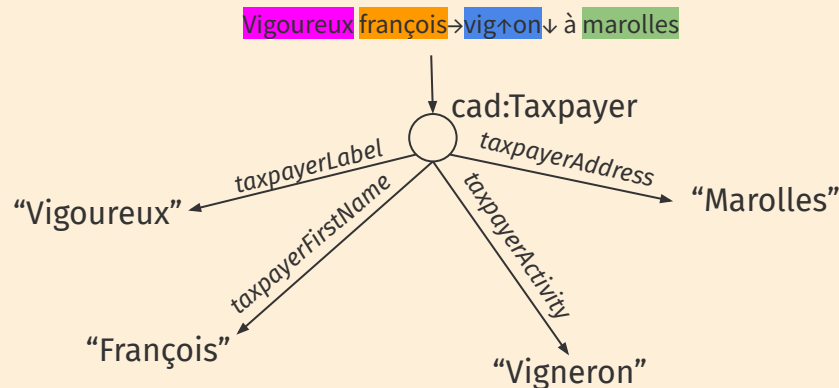
Semantic table interpretation

- CPA, CTA and row-type are fixed thanks to the IE annotation model
- CEA/CNEA and Row-to-Instance Annotation :

CANTON, TRIAGES, ou Lieux dits.	NUMÉROS DU PLAN provisoires définitifs	NOMS, PROFESSIONS, DEMEURES des Propriétaires ou Usufruitiers.	NATURES des Propriétés.	CONTENANCE par Propriété.	CONTENANCE par nature de culture.
				a. p. m.	a. p. m.
De Village	+ 101.	Vigoureux François Vig. à Marolles	Maison et cour	- 1 60	o. M. de B. ch.
	+ 102.	Vigoureux Jean milhane	Maison et cour	- 1 25	o. M. de B. ch.
	+ 103.	Jean	Vigneron	- 7 90	3
	+ 104.	Burroy, Pierre Garde Comptable	Vigneron	- 1 90	2
	+ 105.	Jean	Maison	- 1 00	o. M. de B. ch.



Detection and Creation of New Entities

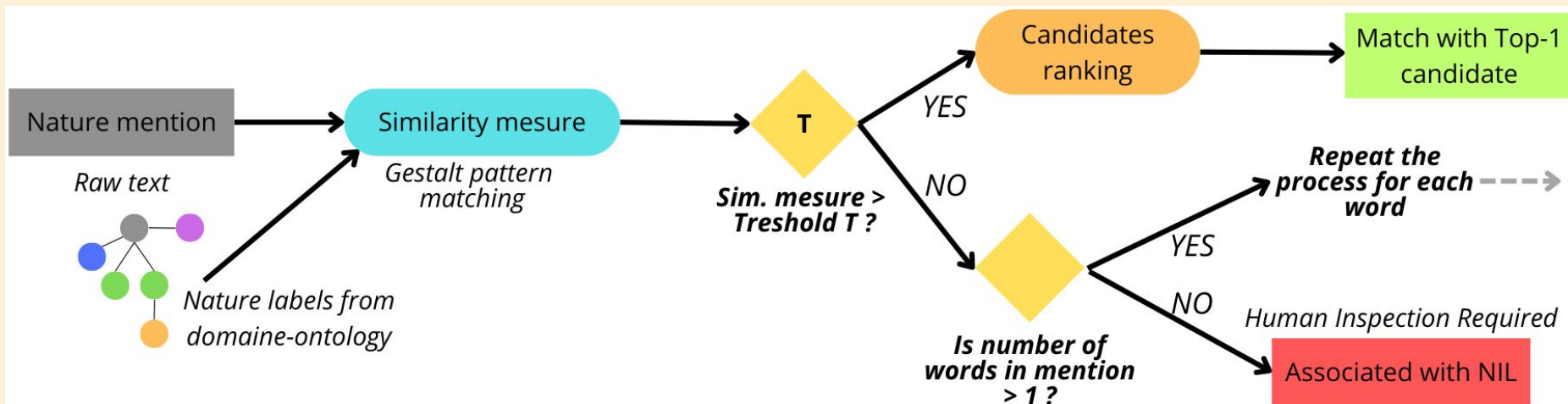


Data collection and preprocessing

- **Registers and metadata collection:**
 - Folders of images structured using archives identifiers
 - Metadata scrapped from archives website + additional archives forms + manual observations in the images:
 - subtype of initial register
 - date of production
 - town
- **Table detection:** performed with YOLOv11 [1] fine-tuned for pages classification
 - 5 classes: *cover pages, main tables, second page of main tables, intermediate summaries, syntheses*
 - 99% accuracy on test set (dataset of 1016 pages)

Cell-Entity Annotation

- Cell-Entity Annotation: “Nature” column
 - Candidates exist in the PeGazUs ontology



Cell-New Entity Annotation

- Creation of new entities: “Taxpayer” and “Address” columns
 - **Candidates don’t exist in the PeGazUs ontology**
 - For Taxpayer: NER is performed with a fined-tuned model to structure the cell value

